
ASSIGNMENTS

Problem set 1

AI504 — Knowledge Representation

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Problem 1

An *interval* is an ordered pair of real numbers (a, b) such that $a < b$. We say that the interval *interleaves* the interval (c, d) in case $a < c < b < d$. Is the “interleaves” relation transitive? If not, what is its transitive closure?

Solution

The relation definition is that

$$\text{if } (x, y) \in R \text{ then } xRy.$$

The interleaves relation is defined as:

$$(a, b)R(c, d) = a < c < b < d.$$

For transitivity

$$(a, b)R(c, d) \wedge (c, d)R(e, f) \Rightarrow (a, b)R(e, f).$$

By the interleaves definition

$$(a, b)R(c, d) = a < c < b < d$$

$$(c, d)R(e, f) = c < e < d < f$$

We wish to prove that $a < e \wedge b < f \wedge e < b$

We can then prove that

$$a < c \wedge c < e \Rightarrow a < e$$

and

$$b < d \wedge d < f \Rightarrow b < f$$

since we only know that

$$c < e < d \wedge c < b < d$$

We cannot say anything about whether $e < b$ is true

So take a counter example

that

$$c < e < d \wedge c < b < d \Rightarrow e = b \vee e < b \vee e > b$$

We can denote the relation closure R^+ as

$$R^+ = \{(e, b)\}$$

Problem 2

Let x and Y be sets, let R and S be binary relations of type $X \times Y$, and let P and Q be subsets of X . Recall that the *forward image* $R(P)$ is the subset of Y obtained by following all R -arrows forwards out of P . (And similarly for all the other forward images in this problem.) For each of the following identities, either prove correct or provide a counterexample. *All proofs are short and all counterexamples are tiny*

(a) $R(P \cup Q) = R(P) \cup R(Q)$

(b) $R(P \cap Q) = R(P) \cap R(Q)$

(c) $(R \cup S)(P) = R(P) \cup S(P)$

(d) $(R \cap S)(P) = R(P) \cap S(P)$

Solution

For this problem

$$R = \{(x, y) \mid \text{some relation}\}$$

(a)

We wish to show that

$$R(P \cup Q) = R(P) \cup R(Q)$$

We know that

$$\forall y \in R(P \cup Q), \exists x \in P \cup Q \mid (x, y) \in R$$

This just means that for every y from the relation on the union of the subsets P and Q there must be some x where the relation R holds.

Then we can show that the y can either come from $R(P)$ or $R(Q)$.

$$\forall y \in R(P) \cup R(Q) : y \in R(P) \vee y \in R(Q)$$

This is the same as $R(P \cup Q) = R(P) \cup R(Q)$

□

(b)

We can apply on $R(P \cap Q) = R(P) \cap R(Q)$

The same applies for

$$\forall y \in R(P \cap Q), \exists x \in P \cap Q \mid (x, y) \in R$$

again the y must come from an x mapped R

But when we do

$$\forall y \in R(P) \cap R(Q) : y \in R(P) \wedge y \in R(Q)$$

We realise that for y to exist P and Q must share an element that both maps to the same y if they don't then **this does not hold**.

Let's do a **counterexample** to properly disprove this

Let set X be a set, and Y be the forward image of relations R_1, S_1 and R_2, S_2

Then let P be a subset of X

Let's check if

$$R(P \cap Q) = R(P) \cap R(Q)$$

When

$$X = \{1, 2\} \text{ and } Y = \{1, 2\} \text{ and } P = \{1\} \subseteq X \text{ and } Q = \{2\} \subseteq X$$

Now let

$$R = \{(1, 2), (2, 2)\}$$

now we can check

$$R(P \cap Q) = \{(1, 2), (2, 2)\}(\{1\} \cap \{2\}) = \emptyset$$

and

$$R(P) \cap R(Q) = \{(1, 2), (2, 2)\}(\{1\}) \cap \{(1, 2), (2, 2)\}(\{2\}) = \{2\} \cap \{2\} = \{2\}$$

Since $\emptyset \neq \{2\}$ this **counters** the statement.

□

(c)

For

$$(R \cup S)(P) = R(P) \cup S(P)$$

We can say that any y must come from an x in P applied by R or S

$$\forall y \in (R \cup S)(P), \exists x \in P \mid (x, y) \in R \text{ or } S$$

Then we can describe $y \in R(P) \cup S(P)$

$$\forall y \in R(P) \cup S(P), \exists x \in P \mid (x, y) \in R \text{ or } S$$

(d)

For

$$(R \cap S)(P) = R(P) \cap S(P)$$

any y must have an x from P which is applied by R and S

$$\forall y \in (R \cap S)(P), \exists x \in P \mid (x, y) \in R \cap S$$

Then showing that x can only be

$$\forall y \in R(P) \cap S(P), \exists x \in P \mid (x, y) \in R \wedge S$$

Let's do a **counterexample** to disprove this

Let set X be a set, and Y be the forward image of relations R_1, S_1 and R_2, S_2

Then let P be a subset of X

Then i want to see if

$$(R_1 \cap S_1)(P) = R_2(P) \cap S_2(P)$$

$$X = \{1, 2\} \text{ and } Y = \{1, 2\} \text{ and } P = \{1, 2\} \subseteq X$$

Now let

$$R_1 = \{(1, 2), (1, 3), (3, 1)\} \quad S_1 = \{(1, 3), (2, 1), (2, 2), (3, 3)\}$$

then

$$R_1 \cap S_1 = \{(1, 3)\}$$

now we can check

$$(R_1 \cap S_1)(P) = \{(1, 3)\}(\{1, 2\}) = \{3\}$$

and

$$\begin{aligned} R_2(P) \cap S_2(P) &= \{(1, 2), (1, 3), (3, 1)\}(\{1, 2\}) \cap \{(1, 3), (2, 1), (2, 2), (3, 3)\}(\{1, 2\}) \\ &= \{1, 2, 3\} \cap \{1, 2, 3\} = \{1, 2, 3\} \end{aligned}$$

Since $\{3\} \neq \{1, 2, 3\}$ this **counters** the statement. □

Problem 3

A *full binary* tree is a nonempty binary tree where each node is a leaf or has exactly two children. Define a function j from full binary trees to rational numbers by recursion as follows: